Lecture 18: Counting I

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1 Rules for Counting

Here are some rules for counting.

1.1 Sum Rule

Rule 1 (Sum rule). Given some list of sets $S_1, S_2, S_3, \ldots, S_k$, where the sets are disjoint, i.e.: $S_i \cap S_j = \emptyset$ for $i \neq j$, the union of all the sets is the sum of the sets, i.e.:

$$\boxed{|S_1 \cup S_2 \cup \cdots \cup S_k| = \sum_{i=1}^{k} |S_i|}$$

1.2 Product Rule

Rule 2 (Product rule). Given some list of sets $S_1, S_2, S_3, \ldots, S_k$, the total number of combinations of a Cartesian product of all the sets is equal to the product of the size of the sets, i.e.:

$$\boxed{|S_1 \times S_2 \times \cdots \times S_k| = \prod_{i=1}^{k} |S_i|}$$

For example, how many binary strings of length 10?

$$\underline{2} \quad \underline{2} \quad \underline{2} \quad \underline{2} \quad \underline{2} \quad \underline{2} \quad \underline{2} \quad \underline{2} \quad \underline{2} \quad \underline{2}$$

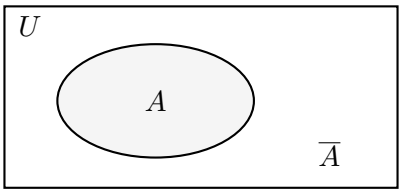
Each slot can use 2 possible digits (0 or 1), so using the product rule, you can have

$$\boxed{2^{10} = 1024.}$$

1.3 Complement Rule

Rule 3 (Complement rule). Given a universal set U, for a set A in U: A is the difference of the universe and the complement of A, i.e.:

$$\boxed{|A| = |U| - |\overline{A}|}$$



For example, how many binary strings of length 10 have at least one 1?

$$|U| = 2^{10}, \qquad |\overline{A}| = 1 \ \text{ (string of 0s)}, \qquad \boxed{|A| = 2^{10} - 1 = 1023.}$$

Counting the strings with at least one 1 directly would mean counting the strings with exactly one 1, exactly two 1s, and so on up to ten 1s. Instead, it is easier to count the strings we don't want, i.e.: the strings with no 1s, and subtract them from the universe.

1.4 Bijection Rule

Rule 4 (Bijection rule). Given some bijection between two sets S and T, the two sets are the same size, i.e.:

$$\boxed{|S| = |T|}$$

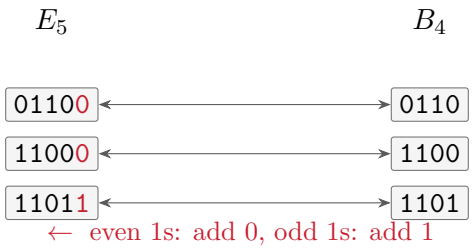
For example, how many bit strings of length 5 have an even number of 1s? Some of these strings are

$$01100, \qquad 11000, \qquad 11011.$$

$$E_5 + O_5 = B_5$$

(E_5 and O_5 partition B_5: every bit string of length 5 has either an even or an odd number of 1s, never both, so the sets are disjoint and together make up all of B_5.)

You can construct a 4 bit long string, because if the 1s are even, you add 0, then if odd, add 1. For example:



$$11011 \in E_5 \longleftrightarrow 1101 \in B_4$$

Each string in E_5 matches with exactly one string in B_4 (remove the last bit), and each string in B_4 matches with exactly one string in E_5 (add the last bit), so this is a bijection between E_5 and

B_4. Using the bijection rule,

$$\boxed{|E_5| = |B_4| = 2^4 = 16.}$$

Since $E_5 + O_5 = B_5$, there are $2^5 - 16 = 16$ strings with an odd number of 1s, i.e.: half of the strings have an even number of 1s.

2 Permutations

No replacement: permutations have no replacement, i.e.: once an element is picked, it can't be picked again.

2.1 Example 1: Permutations of Phone Numbers

Given a phone number with 10 slots and 10 possible digits, i.e.:

__ __ __ __ __ __ __ __ __ __

$$\text{digits} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

(The set of digits is every digit that can go in a slot of the phone number, so each slot starts with 10 possible digits.)

How many ways can you order the digits without repeating?

The first slot can use 10 possible digits, and one possible digit is used, leaving 9 possible digits. The second slot can use 9 possible digits, leaving 8 possible digits. This pattern continues for every slot, i.e.:

10 9 8 7 6 5 4 3 2 1 ← each number represents the size of a subset of digits

We can use the Product Rule to determine the number of ways to order the digits in the phone number (a slot is a subset of digits that contains the elements not used in a previous subset), i.e.:

$$10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \;=\; 10!$$

There are $10! = 3{,}628{,}800$ ways, where

$$\boxed{n! = n \times (n-1) \times (n-2) \times \cdots \times 1.}$$

2.2 Example 2: Permutations of Bank PINs

Given a bank PIN with 4 slots and 10 possible digits, i.e.:

$$__ \quad __ \quad __ \quad __$$

$$\text{digits} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

How many ways can you order the digits without repeating?

The first slot can use 10 possible digits, leaving 9 possible digits. The second slot can use 9 possible digits, leaving 8 possible digits. This continues for every slot, i.e.:

$$\underline{10} \quad \underline{9} \quad \underline{8} \quad \underline{7} \leftarrow \text{using the product rule, the possible combinations is } 10 \times 9 \times 8 \times 7$$

We can say this is equal to

$$\frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

which is equivalent to $\frac{10!}{6!}$.

There are $\frac{10!}{6!}$ ways to order the digits, which we can call permutations of the digits. Generally,

$$\boxed{P(n, k) \text{ or } {}_nP_k = \frac{n!}{(n-k)!}.}$$

2.3 Example 3: Counting Class Positions

A class has 10 students: President, Secretary, Treasurer.

$${}_{10}P_3 = \frac{10!}{7!} = 10 \times 9 \times 8 = \boxed{720}$$

The order matters here, since each position is different (Alice as President and Bob as Secretary is not the same as Bob as President and Alice as Secretary), so we use permutations.

2.4 Example 4: Subsets of a Set of Size 4

Given the set $\{1, 2, 3, 4\}$ of size 4, how many subsets of size 3? E.g.:

$$\underline{1\,2\,3}\,4 \qquad \underline{2\,3\,1}\,4 \qquad \underline{3\,2\,1}\,4$$

If order mattered, picking 3 of the 4 elements gives

$$P(4, 3) = \frac{4!}{(4-3)!} = 4! = 24$$

ways (once the first 3 slots are filled, the last element is the only one left, which is why $P(4,3) = 4!$). But 1 2 3, 2 3 1, and 3 2 1 are all the same subset $\{1,2,3\}$, since sets don't care about the order. Each subset of size 3 shows up 3! times in $P(4,3)$, i.e.:

subset	permutations (counted by $P(4,3)$)					
$\{1,2,3\}$	123	132	213	231	312	321
$\{1,2,4\}$	124	142	214	241	412	421
$\{1,3,4\}$	134	143	314	341	413	431
$\{2,3,4\}$	234	243	324	342	423	432

You are picking the number of combinations $k!$ times, which are the permutations, so

$$C(n,k) \cdot k! = P(n,k), \qquad C(n,k) = \frac{P(n,k)}{k!},$$

written

$$_nC_k \quad \text{or} \quad C(n,k) \quad \text{or} \quad \binom{n}{k}.$$

which is equivalent to

$$\boxed{C(n,k) = \frac{n!}{k!\,(n-k)!}.}$$

So the number of subsets of size 3 is

$$\boxed{C(4,3) = \frac{P(4,3)}{3!} = \frac{24}{6} = 4.}$$

Unlike the examples before it, this example only cares about which elements are picked, not the order they are picked in. Counting this way is called counting *combinations*, which the next section covers.

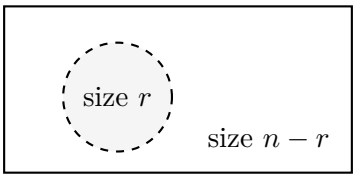
3 Combinations

3.1 Claim: $\binom{n}{r} = \binom{n}{n-r}$

$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!} \qquad \binom{n}{n-r} = \frac{n!}{(n-(n-r))!\,(n-r)!} = \frac{n!}{r!\,(n-r)!}$$

or

$$\text{sets of size } r \;=\; \text{sets of size } (n-r).$$



Each time you pick a set of size r (inside the circle), the elements you don't pick make a set of size $n-r$ (outside the circle), i.e.: the complement. Picking the inside is the same as picking the outside, so this is a bijection between the sets of size r and the sets of size $n-r$. Using the bijection rule, they are the same size.

3.2 Example 1: Counting Combinations of Binary Strings

(Combinations have no replacement, but the order doesn't matter.) How many binary strings of length 9 have exactly 4 1s? You have ${}_9C_4$ of them!

Each binary string matches with the set of slots that have a 1, e.g.:

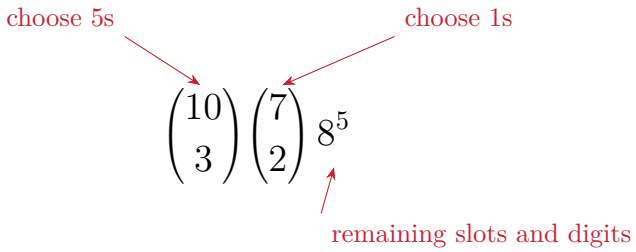
$$0\,1\,1\,0\,0\,1\,0\,1\,0 \longleftrightarrow \{2,3,6,8\}$$

So we are choosing 4 of the 9 slots to put the 1s in, and the rest of the slots get 0s, i.e.:

$${}_9C_4 = \binom{9}{4} = \boxed{126.}$$

3.3 Example 2: Counting Phone Number Combinations

How many phone numbers have 3 5s and 2 1s?



First choose 3 of the 10 slots for the 5s, leaving 7 slots. Then choose 2 of the 7 remaining slots for the 1s, leaving 5 slots. The remaining 5 slots can use 8 possible digits (any digit except 5 and 1). Using the product rule:

$$\binom{10}{3}\binom{7}{2}8^5 = 120 \times 21 \times 32768 = \boxed{82{,}575{,}360.}$$

3.4 Example 3: Counting Combinations of Engineers

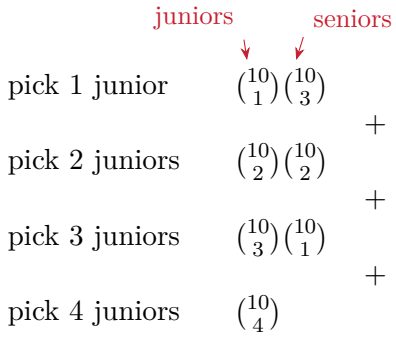
10 senior engineers, 10 junior engineers. Teams of 4 people. How many ways of constructing teams that have at least 1 junior?

All possible teams: $\binom{20}{4}$. Teams of 4 people with no juniors: $\binom{10}{4}$.

Using the complement rule, all possible teams minus the teams with no juniors is the teams with at least 1 junior. Answer:

$$\binom{20}{4} - \binom{10}{4} = 4845 - 210 = \boxed{4635}$$

or, using the sum rule, split the teams by the number of juniors. These cases are disjoint, since a team can't have exactly 1 junior and exactly 2 juniors at the same time:



$$1200 + 2025 + 1200 + 210 = 4635.$$

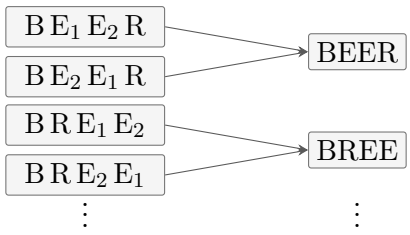
Both ways give the same answer.

4 Permutations with Repetition

4.1 Example 4: Permutations of the Word BEER

Permutations of the word BEER:

$$\mathrm{B\,E_1\,E_2\,R} \longrightarrow 4!$$



$$\frac{4!}{2!} \quad \leftarrow \text{num of repetitions}$$

The subscripts make the two Es different, so there are 4! ways to order $\mathrm{B\,E_1\,E_2\,R}$. But $\mathrm{E_1}$ and $\mathrm{E_2}$ are really the same letter, so $\mathrm{B\,E_1\,E_2\,R}$ and $\mathrm{B\,E_2\,E_1\,R}$ are the same word. Every word is counted 2! times (the number of ways to order the repeated Es), so we divide by 2!:

$$\boxed{\frac{4!}{2!} = 12.}$$

4.2 Example 5: Permutations of a String of Letters

$$A_1\,B_1\,B_2\,B_3\,C\,A_2$$

If all the letters are unique (using subscripts), there are 6! ways to order them. But a word like CABABB is repeated, since the 2 As can be ordered 2! ways and the 3 Bs can be ordered 3! ways. So each word is counted $2!\,3!$ times, i.e.:

$$\text{unique: } 6!, \qquad \text{repeated: } \boxed{\frac{6!}{2!\,3!} = \frac{720}{12} = 60.}$$

General rule. Given an n length string where each letter is repeated $a_1, \ldots, a_k$ times, divide by the number of ways to order each repeated letter:

$$\boxed{\frac{n!}{a_1!\,a_2! \cdots a_k!}}$$

4.3 Example 6: Permutations of the Word MISSISSIPPI

MISSISSIPPI has 11 letters, i.e.:

letter	count
M	1
I	4
S	4
P	2

$$\boxed{\frac{11!}{1!\,4!\,4!\,2!} = 34{,}650}$$

4.4 Proof

Claim. Let $n = 2k$. Then $2^k \mid n!$, i.e. $n! \equiv 0 \pmod{2^k}$.

$$n! = 1 \times 2 \times 3 \times 4 \times \cdots \times 6 \times \cdots \times 2k = (\,\cdots\,) \times \prod_{i=1}^{k} 2i$$

The even numbers $2, 4, 6, \ldots, 2k$ in $n!$ can be written as $2i$ for $i = 1, \ldots, k$. Pulling a 2 out of each one gives

$$\prod_{i=1}^{k} 2i = 2^k \times k!,$$

so $n! = (\cdots) \times 2^k \times k!$, which means

$$\boxed{2^k \mid n!.}$$

(We can also see this using repetition: a string of length $2k$ with k letters, each repeated twice, e.g. AABBCC, has $\frac{(2k)!}{2!\,2!\cdots 2!} = \frac{(2k)!}{2^k}$ permutations. Since this is a number of permutations, it has to be a whole number, so $2^k \mid (2k)!$.)